

THE A -OPTIMAL DESIGN PROBLEM AND THE S -MATRIX CONJECTURE

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ABSTRACT. The S -matrix conjecture was formulated by Sloane and Harwit in 1976, motivated by an A -optimal design problem arising in spectroscopy. It states that if A is a nonsingular real $n \times n$ matrix whose entries lie in $[0, 1]$, then

$$\|A^{-1}\|_F \geq \frac{2n}{n+1}.$$

Moreover, equality holds if and only if A is an S -matrix. In this paper, we prove the conjecture by combining a variational trace inequality in odd dimensions with a sharp range-constrained estimate for the Moore–Penrose inverse of a centered matrix in even dimensions.

1. INTRODUCTION

Optimal experimental design concerns the choice of observations that estimate unknown parameters as accurately as possible under structural, physical, or economic constraints. In a linear model, the quality of a design is encoded by its information matrix, and different scalar summaries of the inverse information matrix lead to different optimality criteria. Among the classical criteria is A -optimality, which minimizes the sum of the variances of the estimated parameters.

Consider the saturated linear model

$$Y = A\theta + \varepsilon, \quad \mathbb{E}(\varepsilon) = 0, \quad \text{Cov}(\varepsilon) = \sigma^2 I_n,$$

where $A \in \mathbb{R}^{n \times n}$ is nonsingular and $\theta \in \mathbb{R}^n$ is unknown. The least-squares estimator is $\hat{\theta} = A^{-1}Y$, and

$$\text{Cov}(\hat{\theta}) = \sigma^2 A^{-1} A^{-\top} = \sigma^2 (A^\top A)^{-1}.$$

Consequently,

$$\mathbb{E} \|\hat{\theta} - \theta\|_2^2 = \sigma^2 \text{tr}((A^\top A)^{-1}) = \sigma^2 \|A^{-1}\|_F^2. \quad (1.1)$$

Thus, in the saturated setting, minimizing the Frobenius norm of A^{-1} is exactly the A -optimal design problem. Equivalently, it minimizes the total mean squared error of the least-squares estimator; after division by n , it minimizes the average variance of its coordinates.

For $A = (a_{ij}), B = (b_{ij}) \in \mathbb{R}^{n \times n}$, the Frobenius inner product is defined by

$$\langle A, B \rangle := \text{tr}(B^\top A) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ij}.$$

The induced Frobenius norm is $\|A\|_F := \sqrt{\langle A, A \rangle} = \left(\sum_{i=1}^n \sum_{j=1}^n a_{ij}^2 \right)^{1/2}$.

The signed design problem originates in Hotelling's classical work on chemical-balance weighing experiments in 1944 [Hot44]. In such an experiment, an object may be placed

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on either side of the balance, so the entries of the design matrix $A = (a_{ij})$ naturally take both positive and negative values. More generally, suppose that

$$-1 \leq a_{ij} \leq 1, \quad 1 \leq i, j \leq n.$$

The Cauchy–Schwarz inequality gives

$$n^2 = \langle A^\top, A^{-1} \rangle^2 \leq \|A^\top\|_F^2 \|A^{-1}\|_F^2 \leq n^2 \|A^{-1}\|_F^2.$$

Consequently,

$$\|A^{-1}\|_F \geq 1. \quad (1.2)$$

Equality holds if and only if A is a *Hadamard matrix*, that is, $A \in \{-1, 1\}^{n \times n}$, and $A^\top A = nI_n$. In this case,

$$\text{Cov}(\hat{\theta}) = \frac{\sigma^2}{n} I_n,$$

so all parameters are estimated with the same variance, and their estimation errors are uncorrelated. Thus, Hadamard matrices provide optimal signed weighing designs; see also [HS79].

The admissible entries of a design matrix, however, depend on the underlying experimental mechanism. In coded spectroscopy, an aperture or spectral channel is typically closed, open, or partially transmitting. Negative transmission coefficients are therefore not physically available, and the natural constraint becomes

$$0 \leq a_{ij} \leq 1, \quad 1 \leq i, j \leq n. \quad (1.3)$$

This nonnegativity constraint fundamentally changes the design problem: For $n > 1$, this nonnegativity constraint excludes Hadamard matrices; indeed, no nonnegative matrix can attain equality in the signed bound (1.2). Determining the sharp lower bound for $\|A^{-1}\|_F$ under (1.3) leads to the *S*-matrix conjecture.

The corresponding problem under the nonnegativity constraint (1.3) arose in the study of masks for Hadamard-transform spectrometers and image scanners. It was formulated by Sloane and Harwit [SH76] and developed further in [HS79, Slo79]. The candidate extremizers are obtained from Hadamard matrices. Let $\mathbf{1} \in \mathbb{R}^n$ be the all-ones vector. Let H be a normalized Hadamard matrix of order $n+1$, written in the form

$$H = \begin{pmatrix} 1 & \mathbf{1}^\top \\ \mathbf{1} & C \end{pmatrix}, \quad C \in \{-1, 1\}^{n \times n}.$$

The matrix

$$S = \frac{1}{2}(\mathbf{1}\mathbf{1}^\top - C)$$

is a 0–1 matrix and is called an *S*-matrix. Clearly, the existence of an *S*-matrix of order n is equivalent to the existence of a Hadamard matrix of order $n+1$. Since a Hadamard matrix of order $N > 2$ can exist only when N is divisible by 4, a nontrivial *S*-matrix can exist only when

$$n \equiv 3 \pmod{4};$$

the exceptional case $n = 1$ corresponds to the Hadamard matrix of order 2. The famous Hadamard conjecture [Zha08, Conjecture 1] asserts that a Hadamard matrix of order $4m$ exists for every positive integer m , or equivalently, that an *S*-matrix of order $4m - 1$ exists for every positive integer m . Hadamard matrices are currently known to exist for every admissible order at most 664, while 668 is the smallest unresolved order. More specifically, among the admissible orders not exceeding 1208, the only unresolved cases are

$$668, \quad 716, \quad 892, \quad 1132;$$

see [CP25].

Equivalently, a matrix $S \in \{0, 1\}^{n \times n}$ is an S -matrix if and only if it satisfies either, and hence both, of the Gram identities

$$S^\top S = \frac{n+1}{4}(I_n + \mathbf{1}\mathbf{1}^\top), \quad \text{or equivalently} \quad SS^\top = \frac{n+1}{4}(I_n + \mathbf{1}\mathbf{1}^\top). \quad (1.4)$$

This characterization is standard in the theory of Hadamard designs; see [CK07, Sec. V.1]. The two Gram identities are also recorded explicitly in [KM08, p. 51].

Since $(I_n + \mathbf{1}\mathbf{1}^\top)^{-1} = I_n - \frac{1}{n+1}\mathbf{1}\mathbf{1}^\top$, we obtain

$$\|S^{-1}\|_F^2 = \text{tr}((S^\top S)^{-1}) = \frac{4n^2}{(n+1)^2}, \quad \|S^{-1}\|_F = \frac{2n}{n+1}. \quad (1.5)$$

Motivated by this construction, Sloane and Harwit [SH76] conjectured that the value in (1.5) is the universal lower bound for every nonsingular matrix satisfying (1.3), and that equality characterizes the S -matrices. This assertion is now known as the *S-matrix conjecture*, which was explicitly recorded in Zhan's remarkable problem collection, “*Open Problems in Matrix Theory*”, as [Zha08, Conjecture 11]. The main result of this paper is to prove this conjecture.

Theorem 1.1 (*S-matrix conjecture*). *Let $A \in \mathbb{R}^{n \times n}$ be nonsingular with $0 \leq a_{ij} \leq 1$, $1 \leq i, j \leq n$. Then*

$$\|A^{-1}\|_F \geq \frac{2n}{n+1}. \quad (1.6)$$

Equality holds if and only if A is an S -matrix.

The S -matrix conjecture has a long history in optimal design and matrix analysis, which may be summarized chronologically as follows:

- In 1960, Kiefer and Wolfowitz [KW60] established an equivalence theorem for optimal experimental design. Kiefer [Kie74] subsequently developed the general equivalence theory in 1974. These results later provided the principal design-theoretic framework for studying the S -matrix problem. The conjecture itself was formulated by Sloane and Harwit [SH76] in 1976 and was developed further in [HS79, Slo79].
- In 1987, Cheng [Che87] applied the Kiefer–Wolfowitz equivalence theory to prove the conjectured bound when n is odd. For even n , he obtained the weaker estimate

$$\|A^{-1}\|_F > \frac{2\sqrt{n^2 - 2n + 2}}{n}.$$

- In 2010, Zou, Jiang, and Hu [ZJH10] established the conjectured bound for a spectrally restricted class of positive definite matrices.
- In 2012, Zou [Zou12] proved the conjectured estimate under an additional Frobenius-norm hypothesis. In the same year, Hu [Hu12] established the conjecture for positive-definite matrices.
- In 2013, Drnovšek [Drm13] gave a short proof of Cheng's odd-dimensional result and even-dimensional bound using the Cauchy–Schwarz inequality and elementary optimization.
- In 2025, Frankel and Urschel [FU25] proved the conjecture for every $n \geq 1000$. Their proof is based on a quantitative near-integrality argument.

(1.6) is equivalent to

$$\text{tr}((A^\top A)^{-1}) \geq \frac{4n^2}{(n+1)^2}. \quad (1.7)$$

Combining (1.1) and (1.7), we obtain

$$\mathbb{E} \|\hat{\theta} - \theta\|_2^2 \geq \frac{4n^2}{(n+1)^2} \sigma^2,$$

and hence

$$\frac{1}{n} \sum_{j=1}^n \text{Var}(\hat{\theta}_j) \geq \frac{4n}{(n+1)^2} \sigma^2.$$

Whenever an S -matrix of order n exists, it is therefore an A -optimal saturated design among all matrices with entries in $[0, 1]$. In that case, the binary design problem and its continuous relaxation have the same optimal value. If no S -matrix of order n exists, Theorem 1.1 shows that the lower bound (1.6) is not attained; it does not determine the exact optimal value or compare the best binary design with the best fractional design in that dimension.

At the level of the two universal lower bounds proved above, the signed problem satisfies

$$\|A^{-1}\|_F^2 \geq 1$$

by (1.2), whereas the nonnegative problem satisfies

$$\|A^{-1}\|_F^2 \geq \frac{4n^2}{(n+1)^2},$$

whose right-hand side converges to 4 as $n \rightarrow \infty$. This is a comparison of the two lower-bound expressions, not necessarily of two simultaneously attainable optima in the same dimension.

Theorem 1.1 also admits the following homogeneous formulation.

Corollary 1.2. *Let $A = (a_{ij}) \in \mathbb{R}^{n \times n}$ be nonsingular and nonnegative. Let $\|A\|_{\max} = \max_{i,j} a_{ij}$. Then*

$$\|A^{-1}\|_F \geq \frac{2n}{n+1} \|A\|_{\max}^{-1}.$$

Equality holds if and only if A is a positive scalar multiple of an S -matrix.

Sketch of the proof. The proof has three components. First, in odd dimensions a fixed matrix Y yields a variational lower inequality for $\text{tr}((AA^\top)^{-1})$. The inequality is sharp exactly on the middle Hamming layer of the cube, and its equality conditions recover Hadamard orthogonality. Second, for even n we prove the centered estimate

$$\|D^\dagger\|_F^2 \geq \frac{n-1}{n+1}$$

for every rank- $(n-1)$ matrix D whose rows sum to zero and have range at most 2. The case $n = 4$ is polyhedral. For $n \geq 6$, a frame-potential functional has no negative minimum: rowwise second variation forces any hypothetical minimizer onto projected vertices of the cube, where parity supplies the missing gap. Finally, an exact formula for A^{-1} separates its centered part from the one-dimensional direction lost under row centering.

Organization of the paper. In Section 2, we give a simple proof of Theorem 1.1 in odd dimensions by combining a variational trace inequality with a sharp maximization problem over the unit cube, and we determine the corresponding equality cases. In Section 3, we establish a range-constrained estimate for the Moore–Penrose inverse of a centered matrix. The case $n = 4$ is treated through a 3-dimensional polyhedral reduction, while the case of even $n \geq 6$ is handled using an inequality for projected sign matrices, a rowwise second-variation argument, and a spectral-variance estimate. In Section 4, we prove the even case in Theorem 1.1 by means of an exact inverse decomposition and a centered energy identity. We also treat the case $n = 2$ directly and complete the proof of Theorem 1.1.

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2. THE ODD CASE

In this section, we give a simple proof of the odd-dimensional case of Theorem 1.1.

We begin with a variational inequality for the inverse trace. The following completion-of-squares inequality will be applied to $G = AA^\top$ with a suitably chosen symmetric matrix Y .

Lemma 2.1. *If $G \in \mathbb{R}^{n \times n}$ is positive definite and $Y \in \mathbb{R}^{n \times n}$ is symmetric, then*

$$\operatorname{tr} G^{-1} \geq 2 \operatorname{tr} Y - \operatorname{tr}(YGY).$$

Equality holds if and only if $Y = G^{-1}$.

Proof. Notice that

$$\operatorname{tr} G^{-1} - 2 \operatorname{tr} Y + \operatorname{tr}(YGY) = \|G^{1/2}(Y - G^{-1})\|_F^2.$$

□

To make the trace inequality effective, we next identify the sharp maximum of the quadratic form induced by the matrix Q over the unit cube. In odd dimensions, the maximizers lie precisely on a single Hamming layer, which will later determine the equality cases.

Let I_n denote the $n \times n$ identity matrix, let $\mathbf{1} \in \mathbb{R}^n$ be the all-ones vector, and put $J_n = \mathbf{1}\mathbf{1}^\top$.

Lemma 2.2. *Suppose that n is odd, and define $Q = I_n - 1/(n+1)J_n$. Then*

$$\max_{x \in [0,1]^n} \|Qx\|_2^2 = \frac{n}{4}.$$

Equality holds if and only if $x \in \{0,1\}^n$ and $\mathbf{1}^\top x = \frac{n+1}{2}$.

Proof. Since $Q^2 = I_n - \frac{n+2}{(n+1)^2}J_n$, we have

$$\|Qx\|_2^2 = \sum_{i=1}^n x_i^2 - \frac{n+2}{(n+1)^2} \left(\sum_{i=1}^n x_i \right)^2. \quad (2.1)$$

The matrix Q^2 is positive definite, so the function in (2.1) is strictly convex. Consequently, any maximizer over the cube must be a cube vertex. At a vertex with k nonzero entries, the value of (2.1) is

$$q(k) = k - \frac{n+2}{(n+1)^2} k^2.$$

Since n is odd, the unique maximizing integer is $k = (n+1)/2$. At a cube vertex, equality occurs exactly when $x \in \{0,1\}^n$ has $(n+1)/2$ nonzero entries, or equivalently, $\mathbf{1}^\top x = (n+1)/2$. Moreover,

$$q\left(\frac{n+1}{2}\right) = \frac{n}{4}.$$

Finally, strict convexity rules out equality at any nonvertex. Indeed, if a nonvertex were written as a nontrivial convex combination of two distinct points of the cube, its value would be strictly smaller than the larger of the two endpoint values. □

We now combine Lemma 2.1 with Lemma 2.2 to prove Theorem 1.1 in odd dimensions. The equality conditions in these two lemmas also provide a complete characterization of the extremizers.

Proposition 2.3. *The conclusion of Theorem 1.1 holds whenever n is odd.*

Proof. Let $G = AA^\top$, and set $Q = I_n - \frac{1}{n+1}J_n$ and $Y = \frac{4}{n+1}Q$. Since $\text{tr } Q = \frac{n^2}{n+1}$, Lemma 2.1 gives

$$\|A^{-1}\|_F^2 = \text{tr } G^{-1} \geq \frac{8n^2}{(n+1)^2} - \frac{16}{(n+1)^2} \|QA\|_F^2. \quad (2.2)$$

Applying Lemma 2.2 to each of the n columns of A yields

$$\|QA\|_F^2 \leq \frac{n^2}{4}. \quad (2.3)$$

Combining (2.2) and (2.3), we obtain

$$\|A^{-1}\|_F^2 \geq \frac{4n^2}{(n+1)^2}. \quad (2.4)$$

It remains to characterize the equality case in (2.4). Suppose that equality holds. Then equality must hold in both (2.2) and (2.3). By Lemma 2.1, equality in (2.2) implies

$$G^{-1} = \frac{4}{n+1}Q.$$

Moreover, by Lemma 2.2, equality in every columnwise estimate used in (2.3) forces A to be a 0–1 matrix, each of whose columns contains exactly $(n+1)/2$ nonzero entries. Since $Q^{-1} = I_n + J_n$, we obtain

$$AA^\top = \frac{n+1}{4}(I_n + J_n).$$

By the equivalent Gram characterization of S -matrices in (1.4), this shows that A is an S -matrix.

Conversely, every S -matrix attains equality by (1.5). $\square$

3. A RANGE-CONSTRAINED PSEUDOINVERSE ESTIMATE

First, we recall the definition of the Moore–Penrose inverse of a matrix; see [BG03, p. 40].

Definition 3.1. Let $M \in \mathbb{R}^{n \times n}$. The *Moore–Penrose inverse* of M , denoted by $M^\dagger$, is the unique matrix in $\mathbb{R}^{n \times n}$ satisfying

$$MM^\dagger M = M, \quad M^\dagger MM^\dagger = M^\dagger,$$

and

$$(MM^\dagger)^\top = MM^\dagger, \quad (M^\dagger M)^\top = M^\dagger M.$$

For $x \in \mathbb{R}^n$, set

$$\text{osc}(x) = \max_j x_j - \min_j x_j.$$

The central result of this section is the following estimate.

Theorem 3.2. *Let $n \geq 4$ be even. Suppose that $D \in \mathbb{R}^{n \times n}$ satisfies $\text{rank } D = n - 1$, $D\mathbf{1} = 0$, and $\text{osc}(d_i) \leq 2$ for all $1 \leq i \leq n$, where $d_i^\top$ denotes the i -th row of D . Then*

$$\|D^\dagger\|_F^2 \geq \frac{n-1}{n+1}.$$

We first describe the common row polytope. Let $P = I_n - \frac{1}{n}J_n$ be the orthogonal projection onto $\mathbf{1}^\perp$, and define

$$\mathcal{K}_n = \{x \in \mathbf{1}^\perp : \text{osc}(x) \leq 2\}.$$

Lemma 3.3. *For every $n \geq 2$,*

$$\mathcal{K}_n = P[-1, 1]^n. \quad (3.1)$$

Moreover,

$$\|x\|_2^2 \leq n \quad (x \in \mathcal{K}_n). \quad (3.2)$$

If n is even, equality in (3.2) holds exactly at the balanced sign vectors, that is, at vectors in $\{-1, 1\}^n \cap \mathbf{1}^\perp$.

Proof. Step 1: $P[-1, 1]^n \subset \mathcal{K}_n$. Let $y \in [-1, 1]^n$. Since

$$Py = y - \frac{1}{n}(\mathbf{1}^\top y)\mathbf{1},$$

the vector Py is obtained from y by subtracting the same constant from every coordinate. Hence, for all j, k ,

$$(Py)_j - (Py)_k = y_j - y_k.$$

It follows that

$$\text{osc}(Py) = \max_{j,k}((Py)_j - (Py)_k) = \max_{j,k}(y_j - y_k) = \text{osc}(y).$$

Since $y \in [-1, 1]^n$, we have

$$\text{osc}(y) \leq 2.$$

Moreover,

$$\mathbf{1}^\top Py = 0,$$

because P is the orthogonal projection onto $\mathbf{1}^\perp$. Therefore $Py \in \mathbf{1}^\perp$ and $\text{osc}(Py) \leq 2$, so

$$Py \in \mathcal{K}_n.$$

As $y \in [-1, 1]^n$ was arbitrary, this proves

$$P[-1, 1]^n \subseteq \mathcal{K}_n.$$

Step 2: $\mathcal{K}_n \subset P[-1, 1]^n$. Conversely, let $x \in \mathcal{K}_n$, and set

$$a = \min_j x_j, \quad b = \max_j x_j, \quad c = \frac{a+b}{2}.$$

Since $x \in \mathcal{K}_n$, we have

$$b - a = \text{osc}(x) \leq 2.$$

For every j ,

$$a \leq x_j \leq b,$$

and therefore

$$-\frac{b-a}{2} = a - c \leq x_j - c \leq b - c = \frac{b-a}{2}.$$

Because $b - a \leq 2$, it follows that

$$-1 \leq x_j - c \leq 1 \quad (1 \leq j \leq n).$$

Hence

$$x - c\mathbf{1} \in [-1, 1]^n.$$

Moreover, $x \in \mathbf{1}^\perp$, so $Px = x$, while $P\mathbf{1} = 0$. Consequently,

$$P(x - c\mathbf{1}) = Px - cP\mathbf{1} = x.$$

Thus x is the projection under P of a vector in $[-1, 1]^n$, and hence

$$x \in P[-1, 1]^n.$$

This proves

$$\mathcal{K}_n \subseteq P[-1, 1]^n.$$

Together with the reverse inclusion established above, we obtain (3.1).

Step 3: norm estimate. Recall $a = \min_j x_j$ and $b = \max_j x_j$. Since $x \perp \mathbf{1}$, we have $a \leq 0 \leq b$. The inequality $(b - x_j)(x_j - a) \geq 0$, summed over j , gives

$$\|x\|_2^2 \leq -nab \leq n \frac{(b-a)^2}{4} \leq n.$$

Equality requires $b - a = 2$, $b = -a = 1$, and every coordinate to equal one of the endpoints. The zero-sum condition then forces equal numbers of 1's and -1 's. Conversely, every balanced sign vector plainly gives equality. $\square$

The proof of Theorem 3.2 has two parts. The case $n = 4$ has a small but exceptional coefficient polytope, which we handle first. The case $n \geq 6$ admits a uniform frame-potential argument.

3.1. The case $n = 4$ in Theorem 3.2. Throughout this subsection, let

$$u = (1, 1, -1, -1)^\top, \quad v = (1, -1, 1, -1)^\top, \quad w = (1, -1, -1, 1)^\top, \quad (3.3)$$

and define

$$\mathcal{P}_4 = \{(a, b, c) \in \mathbb{R}^3 : |a| + |b| \leq 1, |a| + |c| \leq 1, |b| + |c| \leq 1\}. \quad (3.4)$$

We first reduce the problem to a 3-dimensional Gram matrix.

Lemma 3.4. *Let $D \in \mathbb{R}^{4 \times 4}$ satisfy*

$$D\mathbf{1} = 0, \quad \text{rank } D = 3, \quad \text{osc}(d_i) \leq 2 \quad (1 \leq i \leq 4),$$

where $d_i^\top$ is the i -th row of D . Write $d_i = a_i u + b_i v + c_i w$, where u, v, w are defined by (3.3). Let $r_i = (a_i, b_i, c_i)^\top \in \mathbb{R}^3$, let $M \in \mathbb{R}^{4 \times 3}$ have i -th row $r_i^\top$, and put $\Gamma = M^\top M$. Then $r_i \in \mathcal{P}_4$ in (3.4) for every i , the matrix Γ is positive definite, and

$$\|D^\dagger\|_F^2 = \frac{1}{4} \text{tr } \Gamma^{-1}. \quad (3.5)$$

Proof. Since $D\mathbf{1} = 0$, every row d_i belongs to $\mathbf{1}^\perp$. The vectors u, v, w form an orthogonal basis of $\mathbf{1}^\perp$, and each has squared norm 4. Hence each row has a unique representation

$$d_i = a_i u + b_i v + c_i w.$$

For $d = au + bv + cw$, its four coordinates are

$$a + b + c, \quad a - b - c, \quad -a + b - c, \quad -a - b + c.$$

The six quantities obtained by halving the pairwise coordinate differences are

$$a \pm b, \quad a \pm c, \quad b \pm c.$$

Therefore, $\text{osc}(d) \leq 2$ if and only if all six of these quantities have modulus at most 1. Since

$$\max\{|a + b|, |a - b|\} = |a| + |b|,$$

and similarly for the other two pairs, this condition is equivalent to

$$|a| + |b| \leq 1, \quad |a| + |c| \leq 1, \quad |b| + |c| \leq 1.$$

Thus $r_i \in \mathcal{P}_4$.

Let $U \in \mathbb{R}^{3 \times 4}$ be the matrix whose rows are $u^\top, v^\top, w^\top$. Then

$$D = MU, \quad UU^\top = 4I_3,$$

and consequently

$$DD^\top = MUU^\top M^\top = 4MM^\top.$$

The nonzero eigenvalues of $D^\top D$ coincide with those of $DD^\top$, while the nonzero eigenvalues of $MM^\top$ coincide with those of $\Gamma = M^\top M$. Hence the three nonzero eigenvalues of $D^\top D$ are four times the eigenvalues of Γ .

Since $\text{rank } D = 3$, we also have $\text{rank } M = 3$, so Γ is positive definite. Summing the reciprocals of the three nonzero eigenvalues of $D^\top D$ gives

$$\|D^\dagger\|_F^2 = \frac{1}{4} \text{tr } \Gamma^{-1}.$$

□

The next lemma describes the row polytope appearing in Lemma 3.4.

Lemma 3.5. *The polytope $\mathcal{P}_4$ in (3.4) is the convex hull of the six axial points*

$$\pm e_1 = \pm(1, 0, 0), \quad \pm e_2 = \pm(0, 1, 0), \quad \pm e_3 = \pm(0, 0, 1),$$

and the eight half-cube points

$$\left(\pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}\right).$$

In particular,

$$\|r\|_2 \leq 1 \quad (r \in \mathcal{P}_4).$$

Proof. All the listed points belong to $\mathcal{P}_4$. Since $\mathcal{P}_4$ is convex, their convex hull is contained in $\mathcal{P}_4$.

For the reverse inclusion, let $(a, b, c) \in \mathcal{P}_4$. The set $\mathcal{P}_4$ and the collection of listed points are invariant under coordinate permutations and independent sign changes. We may therefore assume that

$$a \geq b \geq c \geq 0.$$

Then

$$\begin{aligned} (a, b, c) &= (a - b)e_1 + (b + c)\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) \\ &\quad + (b - c)\left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right) + (1 - a - b)0. \end{aligned}$$

The coefficients are nonnegative because

$$a \geq b \geq c \geq 0 \quad \text{and} \quad a + b \leq 1,$$

and their sum is equal to 1. Moreover, 0 is the midpoint of any pair of opposite axial points. Hence (a, b, c) belongs to the convex hull of the listed points.

Every axial point has norm 1, and every half-cube point has norm $\sqrt{3}/2$. The convexity of the Euclidean norm therefore gives

$$\|r\|_2 \leq 1 \quad (r \in \mathcal{P}_4).$$

□

We next establish the determinant estimate needed in the spectral argument.

Lemma 3.6. *Let $M \in \mathbb{R}^{4 \times 3}$ have i -th row $r_i^\top$, where $r_1, \dots, r_4 \in \mathcal{P}_4$, and put $\Gamma = M^\top M$. Then*

$$\det \Gamma \leq 2. \tag{3.6}$$

Proof. We first show that each row can be replaced by one of the generating points in Lemma 3.5 without decreasing the determinant.

Fix r_2, r_3, r_4 , and set

$$\Gamma_0 = r_2 r_2^\top + r_3 r_3^\top + r_4 r_4^\top.$$

Regarding the first row as variable, define

$$f(r) = \det(\Gamma_0 + r r^\top).$$

Clearly, $f(r_1) = \det \Gamma$. The rank-one determinant identity in adjugate form gives

$$f(r) = \det \Gamma_0 + r^\top \operatorname{adj}(\Gamma_0) r.$$

Since Γ_0 is positive semidefinite, write

$$\Gamma_0 = O \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3) O^\top, \quad \lambda_i \geq 0,$$

where O is orthogonal. Then

$$\operatorname{adj}(\Gamma_0) = O \operatorname{diag}(\lambda_2 \lambda_3, \lambda_1 \lambda_3, \lambda_1 \lambda_2) O^\top,$$

so $\operatorname{adj}(\Gamma_0)$ is also positive semidefinite. It follows that f is a convex function of r .

Let $\mathcal{V}$ denote the set of generating points listed in Lemma 3.5. Since $\mathcal{P}_4 = \operatorname{conv}(\mathcal{V})$, we may write

$$r_1 = \sum_{k=1}^N \alpha_k v_k, \quad v_k \in \mathcal{V}, \quad \alpha_k \geq 0, \quad \sum_{k=1}^N \alpha_k = 1.$$

By convexity,

$$f(r_1) \leq \sum_{k=1}^N \alpha_k f(v_k) \leq \max_{1 \leq k \leq N} f(v_k).$$

Thus r_1 can be replaced by some $v_k \in \mathcal{V}$ without decreasing $\det \Gamma$. Repeating this argument successively for the remaining three rows shows that it suffices to prove the bound when all four rows belong to $\mathcal{V}$.

Suppose that ℓ of these rows are axial and the remaining $4 - \ell$ are half-cube points. Let ℓ_j be the number of axial rows lying on the j -th coordinate axis. Then

$$\ell_1 + \ell_2 + \ell_3 = \ell,$$

and the j -th diagonal entry of Γ is

$$\Gamma_{jj} = \ell_j + \frac{4 - \ell}{4}.$$

Indeed, an axial row on the j -th axis contributes 1 to Γ_{jj} , whereas every half-cube row contributes $1/4$ to each diagonal entry.

Hadamard's determinant inequality [HJ13, p. 505, Theorem 7.8.1] gives

$$\det \Gamma \leq \prod_{j=1}^3 \Gamma_{jj} = \prod_{j=1}^3 \left(\ell_j + \frac{4 - \ell}{4} \right).$$

For fixed ℓ , this product is maximized when the integers ℓ_1, ℓ_2, ℓ_3 are as evenly distributed as possible. Indeed, if $\ell_j \geq \ell_k + 2$, then transferring one unit from ℓ_j to ℓ_k changes the corresponding two factors by

$$\begin{aligned} & \left(\ell_j - 1 + \frac{4 - \ell}{4} \right) \left(\ell_k + 1 + \frac{4 - \ell}{4} \right) - \left(\ell_j + \frac{4 - \ell}{4} \right) \left(\ell_k + \frac{4 - \ell}{4} \right) \\ &= \ell_j - \ell_k - 1 > 0. \end{aligned}$$

Thus, up to permutation, the maximizing distributions and the resulting bounds are

ℓ	(ℓ_1, ℓ_2, ℓ_3)	$\prod_{j=1}^3 \left(\ell_j + \frac{4-\ell}{4} \right)$
0	(0, 0, 0)	1
1	(1, 0, 0)	$\frac{63}{64}$
2	(1, 1, 0)	$\frac{9}{8}$
3	(1, 1, 1)	$\frac{125}{64}$
4	(2, 1, 1)	2

The largest of these values is 2, which proves (3.6). $\square$

The following lemma handles the case in which the trace of Γ is close to its maximal possible value.

Lemma 3.7. *Let $M \in \mathbb{R}^{4 \times 3}$ have i -th row $r_i^\top$, where $r_1, \dots, r_4 \in \mathcal{P}_4$, and put $\Gamma = M^\top M$. If $\text{tr } \Gamma > \frac{15}{4}$, then*

$$\lambda_{\min}(\Gamma) < \frac{37}{36}. \quad (3.7)$$

Proof. By Lemma 3.5, $\|r_i\|_2^2 \leq 1$. Set $\delta_i = 1 - \|r_i\|_2^2$. Then $\delta_i \geq 0$, and

$$\sum_{i=1}^4 \delta_i = 4 - \text{tr } \Gamma < \frac{1}{4}.$$

In particular,

$$\|r_i\|_2^2 = 1 - \delta_i \geq 1 - \sum_{i=1}^4 \delta_i > \frac{3}{4} \quad (1 \leq i \leq 4).$$

Let p_i be the largest absolute coordinate of r_i . If $p_i \leq 1/2$, then

$$\|r_i\|_2^2 \leq 3p_i^2 \leq \frac{3}{4},$$

which is impossible. Suppose instead that $1/2 < p_i \leq 5/6$. Since $r_i \in \mathcal{P}_4$, each of the other two absolute coordinates is at most $1 - p_i$. Hence

$$\|r_i\|_2^2 \leq p_i^2 + 2(1 - p_i)^2 \leq \frac{3}{4},$$

where the last inequality follows from

$$p_i^2 + 2(1 - p_i)^2 - \frac{3}{4} = 3 \left(p_i - \frac{1}{2} \right) \left(p_i - \frac{5}{6} \right) \leq 0.$$

This is again impossible. Therefore

$$p_i > \frac{5}{6}.$$

The coordinate attaining p_i is unique. Indeed, if two coordinates had absolute value p_i , their absolute values would sum to more than 1, contradicting the definition of $\mathcal{P}_4$. We call this unique coordinate the principal coordinate of r_i .

Put $h_i = 1 - p_i$. The two nonprincipal coordinates of r_i have absolute value at most h_i . Consequently,

$$\delta_i \geq 1 - p_i^2 - 2h_i^2 = h_i(3p_i - 1) \geq \frac{3}{2}h_i.$$

Consequently,

$$\sum_{i=1}^4 h_i \leq \frac{2}{3} \sum_{i=1}^4 \delta_i < \frac{1}{6}.$$

Assign each row to its principal coordinate. If some coordinate, say the j_0 -th, is not assigned to any row, then

$$\Gamma_{j_0 j_0} = \sum_{i=1}^4 (r_i)_{j_0}^2 \leq \sum_{i=1}^4 h_i^2 \leq \left(\sum_{i=1}^4 h_i \right)^2 < \frac{1}{36}.$$

Otherwise, all three coordinates are assigned. Since four rows are assigned to three coordinates, the assignment counts must be $(2, 1, 1)$, up to permutation. Choose a coordinate assigned to exactly one row. The corresponding diagonal entry satisfies

$$\Gamma_{j_0 j_0} \leq 1 + \sum_{i=1}^4 h_i^2 \leq 1 + \left(\sum_{i=1}^4 h_i \right)^2 < \frac{37}{36}.$$

In either case, the Rayleigh–Ritz principle [HJ13, p. 235, Theorem 4.2.2(c)] yields

$$\lambda_{\min}(\Gamma) \leq e_{j_0}^\top \Gamma e_{j_0} = \Gamma_{j_0 j_0} < \frac{37}{36},$$

which proves (3.7). $\square$

We can now prove the case $n = 4$ in Theorem 3.2.

Proposition 3.8. *The conclusion of Theorem 3.2 holds for $n = 4$.*

Proof. Let D satisfy the hypotheses of Theorem 3.2 with $n = 4$, and let Γ be the positive-definite matrix supplied by Lemma 3.4. By (3.5), it is enough to prove that

$$\operatorname{tr} \Gamma^{-1} \geq \frac{12}{5}.$$

Suppose first that $\operatorname{tr} \Gamma \leq 15/4$. The arithmetic–harmonic mean inequality applied to the three eigenvalues of Γ gives

$$\operatorname{tr} \Gamma^{-1} \geq \frac{9}{\operatorname{tr} \Gamma} \geq \frac{12}{5}.$$

It remains to consider $\operatorname{tr} \Gamma > 15/4$. Let $0 < x \leq y \leq z$ be the eigenvalues of Γ . By Lemma 3.7,

$$x < \frac{37}{36},$$

while Lemma 3.6 gives

$$xyz = \det \Gamma \leq 2.$$

Hence

$$yz \leq \frac{2}{x}.$$

It follows that

$$\begin{aligned} \operatorname{tr} \Gamma^{-1} &= \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \\ &\geq \frac{1}{x} + \frac{2}{\sqrt{yz}} \\ &\geq \frac{1}{x} + \sqrt{2x} \\ &> \frac{36}{37} + \sqrt{\frac{37}{18}} \end{aligned}$$

$$> \frac{12}{5}.$$

Here the third inequality is due to the function $s \mapsto s^{-1} + \sqrt{2s}$ is decreasing on $(0, 37/36]$, because $37/36 < 2^{1/3}$. The last displayed inequality may be verified directly.

Thus, in both cases,

$$\operatorname{tr} \Gamma^{-1} \geq \frac{12}{5}.$$

Finally, (3.5) gives

$$\|D^\dagger\|_F^2 = \frac{1}{4} \operatorname{tr} \Gamma^{-1} \geq \frac{3}{5} = \frac{n-1}{n+1},$$

as required. $\square$

3.2. The case $n \geq 6$ in Theorem 3.2. We now prove the case $n \geq 6$ in Theorem 3.2. The proof consists of three steps. We first establish a quantitative estimate for projected sign matrices. We then extend this estimate to the entire row polytope by a second-variation argument. Finally, we convert the resulting spectral-variance estimate into the required bound for the Moore–Penrose inverse.

Lemma 3.9. *Let $n \geq 6$ be even. Let $P = I_n - \frac{1}{n}J_n$ and $R \in \{-1, 1\}^{n \times n}$. Define*

$$D = RP, \quad G = D^\top D, \quad T = \operatorname{tr} G, \quad V = \operatorname{tr}(G^2) - \frac{T^2}{n-1}.$$

Then

$$V \geq \frac{n^2(n-2)}{n-1}(T - (n^2 - 1)). \quad (3.8)$$

Proof. Set

$$\rho = R\mathbf{1}, \quad \delta = \frac{\|\rho\|_2^2}{n}, \quad N = R^\top R.$$

Then

$$G = D^\top D = PR^\top RP = PNP.$$

Since G is positive semidefinite and $G\mathbf{1} = 0$, the Cauchy–Schwarz inequality applied to the eigenvalues of $G|_{\mathbf{1}^\perp}$ gives

$$T^2 \leq (n-1) \operatorname{tr}(G^2).$$

Consequently,

$$V = \operatorname{tr}(G^2) - \frac{T^2}{n-1} \geq 0.$$

We first compute $T = \operatorname{tr} G$:

$$T = \operatorname{tr}(PNP) = \operatorname{tr}(NP) = \operatorname{tr} N - \frac{1}{n}\mathbf{1}^\top N\mathbf{1}.$$

Since $\operatorname{tr} N = \|R\|_F^2 = n^2$ and $\mathbf{1}^\top N\mathbf{1} = \|R\mathbf{1}\|_2^2 = \|\rho\|_2^2 = n\delta$,

$$T = n^2 - \delta. \quad (3.9)$$

If $\delta \geq 1$, then

$$T - (n^2 - 1) = 1 - \delta \leq 0.$$

Since $V \geq 0$, inequality (3.8) follows immediately.

Assume now that $0 \leq \delta < 1$. We next obtain a lower bound for $\operatorname{tr}(N^2)$. Let $R^{(j)}$ denote the j -th column of R . Since $N = R^\top R$,

$$N_{jk} = \left\langle R^{(j)}, R^{(k)} \right\rangle.$$

For $j = k$, every entry of $R^{(j)}$ has modulus 1, and hence

$$N_{jj} = n.$$

Now let $j \neq k$. Suppose that the j -th and k -th columns of R agree in a positions and disagree in b positions. Then $a + b = n$ and

$$N_{jk} = a - b = n - 2b.$$

Because n is even, N_{jk} is an even integer. We claim that

$$N_{jk}^2 \geq -2N_{jk}.$$

Indeed, this is immediate when $N_{jk} \geq 0$. If $N_{jk} < 0$, then the evenness of N_{jk} implies $N_{jk} \leq -2$, and therefore

$$N_{jk}^2 = |N_{jk}|^2 \geq 2|N_{jk}| = -2N_{jk}.$$

Consequently,

$$\text{tr}(N^2) = \sum_{j,k=1}^n N_{jk}^2 = n^3 + \sum_{j \neq k} N_{jk}^2 \geq n^3 - 2 \sum_{j \neq k} N_{jk}. \quad (3.10)$$

Moreover,

$$\sum_{j \neq k} N_{jk} = \sum_{j,k=1}^n N_{jk} - \sum_{j=1}^n N_{jj} = \mathbf{1}^\top N \mathbf{1} - \text{tr} N = n\delta - n^2.$$

Substitution into (3.10) gives

$$\text{tr}(N^2) \geq n^3 + 2n^2 - 2n\delta. \quad (3.11)$$

We also need an upper bound for $\|N\mathbf{1}\|_2$. Since $\rho = R\mathbf{1}$, every entry of ρ is a sum of n numbers in $\{-1, 1\}$. Because n is even, every entry of ρ is an even integer. Let $m = |\{i : \rho_i \neq 0\}|$. Every nonzero entry of ρ has modulus at least 2, so $4m \leq \|\rho\|_2^2 = n\delta$. Thus

$$m \leq \frac{n\delta}{4}. \quad (3.12)$$

For each j ,

$$(N\mathbf{1})_j = (R^\top \rho)_j = \sum_{i: \rho_i \neq 0} R_{ij} \rho_i.$$

The Cauchy–Schwarz inequality and (3.12) give

$$((N\mathbf{1})_j)^2 \leq m \sum_{i: \rho_i \neq 0} \rho_i^2 = m \|\rho\|_2^2 \leq \frac{n^2 \delta^2}{4}.$$

Summing over j , we obtain

$$\|N\mathbf{1}\|_2^2 \leq \frac{n^3 \delta^2}{4}. \quad (3.13)$$

We now pass from N to $G = PNP$. Expanding $P = I_n - J_n/n$ and using cyclicity of the trace yields

$$\begin{aligned} \text{tr}(G^2) &= \text{tr}(NPNP) \\ &= \text{tr}(N^2) - \frac{2}{n} \mathbf{1}^\top N^2 \mathbf{1} + \frac{1}{n^2} (\mathbf{1}^\top N \mathbf{1})^2 \\ &= \text{tr}(N^2) - \frac{2}{n} \|N\mathbf{1}\|_2^2 + \delta^2. \end{aligned} \quad (3.14)$$

Combining (3.11), (3.13), and (3.14), we obtain

$$\text{tr}(G^2) \geq n^3 + 2n^2 - 2n\delta - \left(\frac{n^2}{2} - 1\right) \delta^2. \quad (3.15)$$

Finally, by the definitions of T and V ,

$$(n-1)V = (n-1)\operatorname{tr}(G^2) - T^2.$$

Using (3.9) and (3.15), a direct simplification gives

$$\begin{aligned} & (n-1) \left[V - \frac{n^2(n-2)}{n-1} (T - (n^2-1)) \right] \\ & \geq \delta \left[n(n^2 - 2n + 2) - \left(\frac{n^2(n-1)}{2} - n + 2 \right) \delta \right]. \end{aligned} \quad (3.16)$$

The coefficient

$$\frac{n^2(n-1)}{2} - n + 2$$

is positive. Since $0 \leq \delta < 1$, the expression in square brackets in (3.16) is bounded below by

$$n(n^2 - 2n + 2) - \left(\frac{n^2(n-1)}{2} - n + 2 \right) = \frac{(n-1)(n^2 - 2n + 4)}{2} > 0.$$

The right-hand side of (3.16) is therefore nonnegative, which proves (3.8). $\square$

We isolate the second-variation computation used in the extension from projected sign matrices to arbitrary admissible matrices.

Lemma 3.10. *Let $n \geq 6$, and let $D \in \mathbb{R}^{n \times n}$ satisfy $D\mathbf{1} = 0$. Define*

$$G = D^\top D, \quad T = \operatorname{tr} G,$$

and

$$\Phi(D) = \operatorname{tr}(G^2) - \frac{T^2}{n-1} - \frac{n^2(n-2)}{n-1} (T - (n^2-1)).$$

Fix $x, z \in \mathbf{1}^\perp$, and suppose that $x^\top$ is one of the rows of D . For $\tau \in \mathbb{R}$, let D_τ be obtained from D by replacing the row $x^\top$ with $(x + \tau z)^\top$. Then

$$\begin{aligned} \frac{1}{4} \frac{d^2}{d\tau^2} \Big|_{\tau=0} \Phi(D_\tau) &= z^\top G z + \left(1 - \frac{2}{n-1} \right) (x^\top z)^2 \\ &\quad + \left(\|x\|_2^2 - \frac{T}{n-1} - \frac{n^2(n-2)}{2(n-1)} \right) \|z\|_2^2. \end{aligned} \quad (3.17)$$

Proof. Only one row of D is changed. Therefore

$$D_\tau^\top D_\tau = G + \tau(xz^\top + zx^\top) + \tau^2 zz^\top.$$

If $T_\tau = \operatorname{tr}(D_\tau^\top D_\tau)$, then

$$T_\tau = T + 2\tau x^\top z + \tau^2 \|z\|_2^2.$$

Direct differentiation gives

$$\frac{d^2}{d\tau^2} \Big|_{\tau=0} \operatorname{tr}((D_\tau^\top D_\tau)^2) = 4z^\top G z + 4\|x\|_2^2 \|z\|_2^2 + 4(x^\top z)^2$$

and

$$\frac{d^2}{d\tau^2} \Big|_{\tau=0} T_\tau^2 = 8(x^\top z)^2 + 4T\|z\|_2^2.$$

Also,

$$\frac{d^2}{d\tau^2} \Big|_{\tau=0} T_\tau = 2\|z\|_2^2.$$

Substituting these three identities into the definition of $\Phi(D_\tau)$ gives (3.17). $\square$

We now extend Lemma 3.9 to every matrix whose rows belong to the row polytope.

Lemma 3.11. *Let $n \geq 6$ be even, and define*

$$\mathcal{K}_n = \left\{ x \in \mathbb{R}^n : \mathbf{1}^\top x = 0 \text{ and } \text{osc}(x) \leq 2 \right\}.$$

Let $D \in \mathbb{R}^{n \times n}$ have rows $d_1^\top, \dots, d_n^\top$, where $d_1, \dots, d_n \in \mathcal{K}_n$. Define

$$G = D^\top D, \quad T = \text{tr } G, \quad V = \text{tr}(G^2) - \frac{T^2}{n-1}.$$

Then

$$V \geq \frac{n^2(n-2)}{n-1}(T - (n^2 - 1)). \quad (3.18)$$

Proof. For every matrix $X \in \mathbb{R}^{n \times n}$ whose rows belong to $\mathcal{K}_n$, define

$$G(X) = X^\top X, \quad T(X) = \text{tr } G(X),$$

and

$$V(X) = \text{tr}(G(X)^2) - \frac{T(X)^2}{n-1}.$$

We also define

$$\Phi(X) = V(X) - \frac{n^2(n-2)}{n-1}(T(X) - (n^2 - 1)).$$

For the matrix D in the statement of the lemma, (3.18) is precisely the assertion that

$$\Phi(D) \geq 0.$$

Suppose, for a contradiction, that Φ takes a negative value on the set of matrices whose rows belong to $\mathcal{K}_n$. By Lemma 3.3, the set $\mathcal{K}_n$ is compact. Hence the set of such matrices is compact, being the Cartesian product of n copies of $\mathcal{K}_n$. Since Φ is continuous, it attains its global minimum on this set.

Let X_* be an admissible matrix at which this minimum is attained. By the contradiction assumption,

$$\Phi(X_*) < 0.$$

For convenience, set $G_* = X_*^\top X_*$, $T_* = \text{tr } G_*$, and $V_* = \text{tr}(G_*^2) - \frac{T_*^2}{n-1}$. Let $x_1^\top, \dots, x_n^\top$ denote the rows of X_* . Since every x_i belongs to $\mathcal{K}_n$, Lemma 3.3 gives

$$\|x_i\|_2^2 \leq n \quad (1 \leq i \leq n).$$

Consequently,

$$T_* = \|X_*\|_F^2 = \sum_{i=1}^n \|x_i\|_2^2 \leq n^2.$$

Since $X_* \mathbf{1} = 0$, the matrix G_* annihilates $\mathbf{1}$. If $\lambda_1, \dots, \lambda_{n-1}$ are the eigenvalues of $G_*|_{\mathbf{1}^\perp}$, including zero eigenvalues when necessary, then

$$T_* = \sum_{i=1}^{n-1} \lambda_i, \quad V_* = \sum_{i=1}^{n-1} \left(\lambda_i - \frac{T_*}{n-1} \right)^2.$$

In particular,

$$V_* \geq 0.$$

Since

$$\Phi(X_*) = V_* - \frac{n^2(n-2)}{n-1}(T_* - (n^2 - 1)) < 0,$$

we must have

$$T_* > n^2 - 1.$$

We may therefore write

$$T_* = n^2 - 1 + \varepsilon, \quad 0 < \varepsilon \leq 1. \quad (3.19)$$

We next show that $\varepsilon < 1$. Suppose that $\varepsilon = 1$. Then $T_* = n^2$, and hence

$$\sum_{i=1}^n \|x_i\|_2^2 = n^2.$$

Since every summand is at most n , it follows that

$$\|x_i\|_2^2 = n \quad (1 \leq i \leq n).$$

By the equality statement in Lemma 3.3, every row of X_* is a balanced sign vector. Thus

$$X_* \in \{-1, 1\}^{n \times n} \quad \text{and} \quad X_* \mathbf{1} = 0.$$

Set $P = I_n - \frac{1}{n}J_n$. Since $X_* \mathbf{1} = 0$, we have

$$X_* P = X_*.$$

Therefore, taking $R = X_*$, we may write

$$X_* = RP \quad \text{with} \quad R \in \{-1, 1\}^{n \times n}.$$

Lemma 3.9 then gives

$$\Phi(X_*) \geq 0,$$

contradicting the choice of X_* . Hence

$$0 < \varepsilon < 1. \quad (3.20)$$

We now estimate the largest eigenvalue of $G_*|_{\mathbf{1}^\perp}$. Let $\lambda_{\max} = \max_{1 \leq i \leq n-1} \lambda_i$ and set $\eta = \lambda_{\max} - T_*/(n-1)$. The remaining $n-2$ deviations from the mean $T_*/(n-1)$ have sum $-\eta$. By the Cauchy-Schwarz inequality, the sum of their squares is at least $\eta^2/(n-2)$. It follows that

$$V_* \geq \eta^2 + \frac{\eta^2}{n-2} = \frac{n-1}{n-2} \eta^2.$$

Consequently,

$$\lambda_{\max} \leq \frac{T_*}{n-1} + \sqrt{\frac{n-2}{n-1}} V_*.$$

Since $\Phi(X_*) < 0$, equation (3.19) gives

$$V_* < \frac{n^2(n-2)}{n-1} \varepsilon.$$

Therefore,

$$\lambda_{\max} < \frac{T_* + n(n-2)\sqrt{\varepsilon}}{n-1}. \quad (3.21)$$

We next prove that every row of X_* is an extreme point of $\mathcal{K}_n$. Fix $x \in \mathcal{K}_n$ such that $x^\top$ is a row of X_* , and suppose that x is not an extreme point of $\mathcal{K}_n$. Then x lies in the relative interior of a nontrivial line segment contained in $\mathcal{K}_n$. After rescaling the direction of this segment, there exists a nonzero vector $z \in \mathbf{1}^\perp$ such that

$$x + \tau z \in \mathcal{K}_n \quad (-1 \leq \tau \leq 1).$$

For $-1 \leq \tau \leq 1$, let X_τ be obtained from X_* by replacing the row $x^\top$ with $(x + \tau z)^\top$, and define

$$\phi(\tau) = \Phi(X_\tau).$$

Every X_τ is admissible. Since X_* is a global minimizer of Φ , the function ϕ has a local minimum at $\tau = 0$. Hence

$$\phi''(0) \geq 0. \quad (3.22)$$

On the other hand, applying Lemma 3.10 to X_* gives

$$\begin{aligned} \frac{1}{4}\phi''(0) &= z^\top G_* z + \left(1 - \frac{2}{n-1}\right) (x^\top z)^2 \\ &\quad + \left(\|x\|_2^2 - \frac{T_*}{n-1} - \frac{n^2(n-2)}{2(n-1)}\right) \|z\|_2^2. \end{aligned}$$

Since $z \in \mathbf{1}^\perp$, the Rayleigh–Ritz principle [HJ13, p. 235, Theorem 4.2.2(c)] gives

$$z^\top G_* z \leq \lambda_{\max} \|z\|_2^2.$$

Moreover,

$$(x^\top z)^2 \leq \|x\|_2^2 \|z\|_2^2,$$

and Lemma 3.3 gives

$$\|x\|_2^2 \leq n.$$

Since $1 - 2/(n-1) > 0$, we obtain

$$\begin{aligned} \frac{1}{4}\phi''(0) &\leq \left[\lambda_{\max} + \left(2 - \frac{2}{n-1}\right) \|x\|_2^2 - \frac{T_*}{n-1} - \frac{n^2(n-2)}{2(n-1)} \right] \|z\|_2^2 \\ &\leq \left[\lambda_{\max} + \frac{2n(n-2)}{n-1} - \frac{T_*}{n-1} - \frac{n^2(n-2)}{2(n-1)} \right] \|z\|_2^2. \end{aligned}$$

Using (3.21), we find that

$$\frac{1}{4}\phi''(0) < \frac{n(n-2)}{n-1} \left(2 + \sqrt{\varepsilon} - \frac{n}{2}\right) \|z\|_2^2.$$

If $n \geq 8$, then, by (3.20),

$$2 + \sqrt{\varepsilon} - \frac{n}{2} < 3 - \frac{n}{2} \leq -1.$$

If $n = 6$, then

$$2 + \sqrt{\varepsilon} - \frac{n}{2} = \sqrt{\varepsilon} - 1 < 0.$$

Thus, in every case,

$$\phi''(0) < 0.$$

This contradicts (3.22). We conclude that every row of X_* is an extreme point of $\mathcal{K}_n$.

It remains to identify the form of these extreme points. By Lemma 3.3,

$$\mathcal{K}_n = P[-1, 1]^n, \quad P = I_n - \frac{1}{n}J_n.$$

Let x be an extreme point of $\mathcal{K}_n$. Since $\mathcal{K}_n$ is a polytope, there exists a linear functional ℓ that is uniquely maximized over $\mathcal{K}_n$ at x .

The linear functional $s \mapsto \ell(Ps)$ attains its maximum over the cube $[-1, 1]^n$ at some vertex $s \in \{-1, 1\}^n$. Thus Ps maximizes ℓ over $\mathcal{K}_n$. By the uniqueness of the maximizer,

$$x = Ps.$$

Applying this argument to every row of X_* , choose sign vectors $s_1, \dots, s_n \in \{-1, 1\}^n$ such that the i -th row of X_* is $(Ps_i)^\top$. Let $R \in \{-1, 1\}^{n \times n}$ be the matrix whose i -th row is $s_i^\top$. Since $P = P^\top$, the i -th row of RP is

$$s_i^\top P = (Ps_i)^\top.$$

Therefore,

$$X_* = RP.$$

Lemma 3.9 now gives

$$\Phi(X_*) \geq 0,$$

contradicting $\Phi(X_*) < 0$. Hence Φ is nonnegative on the set of admissible matrices. In particular,

$$\Phi(D) \geq 0$$

for the matrix D in the statement, which is exactly (3.18). $\square$

The next elementary lemma converts spectral variance into a lower bound for the sum of reciprocal eigenvalues.

Lemma 3.12. *Let $r \geq 2$, let $\lambda_1, \dots, \lambda_r > 0$, and define*

$$T = \sum_{i=1}^r \lambda_i, \quad \bar{\lambda} = \frac{T}{r}, \quad V = \sum_{i=1}^r (\lambda_i - \bar{\lambda})^2.$$

Then

$$\sum_{i=1}^r \frac{1}{\lambda_i} \geq \frac{r^2}{T} + \frac{V}{\bar{\lambda}^2 \left(\bar{\lambda} + \sqrt{\frac{r-1}{r} V} \right)}. \quad (3.23)$$

Proof. Let $\lambda_{\max} = \max_{1 \leq i \leq r} \lambda_i$ and set $d = \lambda_{\max} - \bar{\lambda}$. The remaining $r - 1$ deviations from $\bar{\lambda}$ have sum $-d$. The Cauchy–Schwarz inequality therefore gives

$$V \geq d^2 + \frac{d^2}{r-1} = \frac{r}{r-1} d^2.$$

Thus

$$\lambda_{\max} \leq \bar{\lambda} + \sqrt{\frac{r-1}{r} V}. \quad (3.24)$$

For every $\lambda > 0$, the following identity is exact:

$$\frac{1}{\lambda} = \frac{1}{\bar{\lambda}} - \frac{\lambda - \bar{\lambda}}{\bar{\lambda}^2} + \frac{(\lambda - \bar{\lambda})^2}{\bar{\lambda}^2 \lambda}.$$

Summing this identity over i and using

$$\sum_{i=1}^r (\lambda_i - \bar{\lambda}) = 0,$$

we obtain

$$\sum_{i=1}^r \frac{1}{\lambda_i} = \frac{r^2}{T} + \sum_{i=1}^r \frac{(\lambda_i - \bar{\lambda})^2}{\bar{\lambda}^2 \lambda_i}. \quad (3.25)$$

Since $\lambda_i \leq \lambda_{\max}$,

$$\sum_{i=1}^r \frac{(\lambda_i - \bar{\lambda})^2}{\bar{\lambda}^2 \lambda_i} \geq \frac{V}{\bar{\lambda}^2 \lambda_{\max}}.$$

Combining this estimate with (3.24) and (3.25) proves (3.23). $\square$

We can now complete the proof for the case $n \geq 6$ in Theorem 3.2.

Proposition 3.13. *The conclusion of Theorem 3.2 holds for all even $n \geq 6$.*

Proof. Let D satisfy the hypotheses of Theorem 3.2. Set $r = n - 1$, $G = D^\top D$, and let $\lambda_1, \dots, \lambda_r > 0$ be the nonzero eigenvalues of G . Since $\text{rank } D = r$, $\|D^\dagger\|_F^2 = \sum_{i=1}^r 1/\lambda_i$. Define

$$T = \sum_{i=1}^r \lambda_i = \text{tr } G = \|D\|_F^2 \quad \text{and} \quad V = \sum_{i=1}^r \left(\lambda_i - \frac{T}{r} \right)^2.$$

By Lemma 3.3,

$$\|d_i\|_2^2 \leq n \quad (1 \leq i \leq n),$$

and hence

$$T \leq n^2.$$

Suppose first that $T \leq n^2 - 1 = r(n + 1)$. The arithmetic–harmonic mean inequality gives

$$\|D^\dagger\|_F^2 = \sum_{i=1}^r \frac{1}{\lambda_i} \geq \frac{r^2}{T} \geq \frac{r}{n+1}.$$

It remains to consider $T > n^2 - 1$. Write $T = n^2 - 1 + \varepsilon$, where $0 < \varepsilon \leq 1$. Since $D\mathbf{1} = 0$ and $\text{rank } D = r = n - 1$, the spectrum of G is

$$0, \lambda_1, \dots, \lambda_r.$$

Hence

$$V = \text{tr}(G^2) - \frac{T^2}{n-1}.$$

Moreover, the hypotheses on the rows of D imply that they belong to $\mathcal{K}_n$. Therefore Lemma 3.11 gives

$$V \geq \frac{n^2(n-2)}{r} \varepsilon. \quad (3.26)$$

Set $\bar{\lambda} = T/r$. Lemma 3.12 gives

$$\|D^\dagger\|_F^2 \geq \frac{r^2}{T} + \frac{V}{\bar{\lambda}^2 \left(\bar{\lambda} + \sqrt{\frac{r-1}{r}} V \right)}. \quad (3.27)$$

For fixed $\bar{\lambda} > 0$, the function

$$v \mapsto \frac{v}{\bar{\lambda} + \sqrt{\frac{r-1}{r}} v}$$

is increasing on $[0, \infty)$. Indeed, for $v > 0$, its derivative is

$$\frac{\bar{\lambda} + \frac{1}{2} \sqrt{\frac{r-1}{r}} v}{\left(\bar{\lambda} + \sqrt{\frac{r-1}{r}} v \right)^2} > 0.$$

We may therefore substitute the lower bound (3.26) into the second term of (3.27). Since $r = n - 1$,

$$\sqrt{\frac{r-1}{r}} \cdot \frac{n^2(n-2)}{r} \varepsilon = \frac{n(n-2)}{r} \sqrt{\varepsilon}.$$

It follows that

$$\frac{V}{\bar{\lambda}^2 \left(\bar{\lambda} + \sqrt{\frac{r-1}{r}} V \right)} \geq \frac{n^2(n-2)\varepsilon r^2}{T^2 (T + n(n-2)\sqrt{\varepsilon})}. \quad (3.28)$$

The difference between the desired bound and the first term in (3.27) is

$$\frac{r}{n+1} - \frac{r^2}{T} = \frac{r\varepsilon}{(n+1)T}. \quad (3.29)$$

Therefore, it suffices to prove that

$$(n+1)n^2(n-2)r \geq T(T+n(n-2)\sqrt{\varepsilon}). \quad (3.30)$$

Since $T \leq n^2$ and $\varepsilon \leq 1$,

$$\begin{aligned} T(T+n(n-2)\sqrt{\varepsilon}) &\leq n^2(n^2+n(n-2)) \\ &= 2n^3(n-1) \\ &= 2n^3r. \end{aligned}$$

On the other hand,

$$(n+1)n^2(n-2)r \geq 2n^3r,$$

because

$$(n+1)(n-2) \geq 2n \quad (n \geq 4).$$

This proves (3.30). Combining (3.27), (3.28), and (3.29), we obtain

$$\|D^\dagger\|_F^2 \geq \frac{r}{n+1} = \frac{n-1}{n+1}.$$

□

Proof of Theorem 3.2. For $n = 4$, the conclusion follows from Proposition 3.8. For every even $n \geq 6$, it follows from Proposition 3.13. □

4. THE EVEN CASE

In this section, we prove the even-dimensional case of Theorem 1.1.

With the centered estimate in Theorem 3.2 available, it remains to recover the direction annihilated by row centering. The following exact decomposition isolates that one-dimensional contribution.

Lemma 4.1. *Let $A \in \mathbb{R}^{n \times n}$ be nonsingular, put*

$$P = I_n - \frac{1}{n}J_n, \quad \mu = \frac{1}{n}A\mathbf{1}, \quad D = 2AP,$$

and let u be a unit vector spanning $\ker D^\top$. Define $t = u^\top \mu$, $w = \mu - tu$. Then $t \neq 0$, and

$$A^{-1} = 2D^\dagger + \left(\frac{1}{nt}\mathbf{1} - \frac{2}{t}D^\dagger w \right) u^\top. \quad (4.1)$$

Moreover,

$$\|A^{-1}\|_F^2 = 4\|D^\dagger\|_F^2 + \frac{1}{nt^2} + \frac{4}{t^2}\|D^\dagger w\|_2^2. \quad (4.2)$$

Proof. Since A is invertible and P has rank $n-1$, D has rank $n-1$, $\ker D = \text{span}\{\mathbf{1}\}$, and $\text{range } D = u^\perp$. If $t = 0$, then $\mu \in u^\perp = \text{range } D$, and every column of $A = 1/2D + \mu\mathbf{1}^\top$ would lie in $\text{range } D$, contradicting the nonsingularity of A . Thus $t \neq 0$, and $w \in \text{range } D$.

The standard Moore–Penrose identities in the present rank-one defect setting are

$$D^\dagger D = P, \quad DD^\dagger = I_n - uu^\top, \quad D^\dagger u = 0.$$

Multiplying the right-hand side of (4.1) by $A = D/2 + \mu\mathbf{1}^\top$, and using $D^\dagger \mu = D^\dagger w$, gives

$$P + 2D^\dagger w\mathbf{1}^\top + \frac{1}{n}J_n - 2D^\dagger w\mathbf{1}^\top = I_n.$$

Thus the displayed matrix is a left inverse of A . Since A is square and nonsingular, this left inverse equals A^{-1} , proving (4.1).

Finally,

$$\text{range}(D^\dagger) = \text{range}(D^\top) = (\ker D)^\perp = \mathbf{1}^\perp,$$

so $D^\dagger w \perp \mathbf{1}$. Also $D^\dagger u = 0$. Hence, for every $z \in \mathbb{R}^n$,

$$\begin{aligned}\langle D^\dagger, zu^\top \rangle &= \text{tr}\left((zu^\top)^\top D^\dagger\right) \\ &= \text{tr}\left(uz^\top D^\dagger\right) = z^\top D^\dagger u = 0.\end{aligned}$$

Moreover, since u is a unit vector,

$$\begin{aligned}\langle \mathbf{1}u^\top, (D^\dagger w)u^\top \rangle &= \text{tr}\left(u(D^\dagger w)^\top \mathbf{1}u^\top\right) \\ &= (u^\top u)\mathbf{1}^\top D^\dagger w = 0.\end{aligned}$$

Thus the three terms in (4.1) are pairwise orthogonal with respect to the Frobenius inner product. Finally,

$$\|\mathbf{1}u^\top\|_F^2 = n, \quad \|(D^\dagger w)u^\top\|_F^2 = \|D^\dagger w\|_2^2,$$

and therefore (4.2) follows. $\square$

The next identity measures simultaneously the loss under centering and the distance from the binary cube.

Lemma 4.2. *Assume that $0 \leq a_{ij} \leq 1$, and use the notation of Lemma 4.1. Put*

$$T = \|D\|_F^2, \quad E = \sum_{i,j} a_{ij}(1 - a_{ij}), \quad q = \left\| \mu - \frac{1}{2}\mathbf{1} \right\|_2^2, \quad x = 2\sqrt{nq}.$$

Then

$$n^2 - T = x^2 + 4E. \tag{4.3}$$

Furthermore,

$$\|\mu\|_2 \leq \frac{n+x}{2\sqrt{n}}. \tag{4.4}$$

Proof. Since $D = 2(A - \mu\mathbf{1}^\top)$, rowwise expansion gives

$$T = 4 \sum_{i,j} a_{ij}^2 - 4n\|\mu\|_2^2.$$

On the other hand,

$$\begin{aligned}4nq + 4E &= 4n\|\mu\|_2^2 - 4n\mathbf{1}^\top \mu + n^2 + 4 \sum_{i,j} a_{ij} - 4 \sum_{i,j} a_{ij}^2 \\ &= n^2 - T,\end{aligned}$$

because $\sum_{i,j} a_{ij} = n\mathbf{1}^\top \mu$. This proves (4.3). The triangle inequality gives

$$\|\mu\|_2 \leq \frac{\sqrt{n}}{2} + \sqrt{q} = \frac{n+x}{2\sqrt{n}},$$

which is (4.4). $\square$

We can now prove strictness in every even dimension at least four. The two cases below are divided at $x = 1$, exactly where the centered estimate and the elementary singular-value estimate meet.

Proposition 4.3. *Let $n \geq 4$ be even. Under the hypotheses of Theorem 1.1,*

$$\|A^{-1}\|_F > \frac{2n}{n+1}.$$

Proof. Use the notation of Lemmas 4.1 and 4.2. Since $P\mathbf{1} = 0$, we have $D\mathbf{1} = 0$. If $a_i^\top$ and $d_i^\top$ are the i -th rows of A and D , respectively, then subtracting the row mean does not change oscillation, and hence

$$\text{osc}(d_i) = 2\text{osc}(a_i) \leq 2.$$

Moreover, since A is nonsingular and $\text{rank } P = n - 1$,

$$\text{rank } D = \text{rank}(2AP) = n - 1.$$

Therefore Theorem 3.2 applies.

Suppose first that $x < 1$. By (4.2), Theorem 3.2, $|t| \leq \|\mu\|_2$, and (4.4),

$$\begin{aligned} \|A^{-1}\|_F^2 &\geq 4\frac{n-1}{n+1} + \frac{1}{n\|\mu\|_2^2} \\ &> 4\frac{n-1}{n+1} + \frac{4}{(n+1)^2} = \frac{4n^2}{(n+1)^2}. \end{aligned}$$

Now assume $x \geq 1$. Since $E \geq 0$ and $T = \|D\|_F^2 > 0$, the identity

$$T = n^2 - x^2 - 4E$$

implies $1 \leq x < n$. The arithmetic-harmonic mean inequality applied to the $n - 1$ nonzero squared singular values of D gives

$$\|D^\dagger\|_F^2 \geq \frac{(n-1)^2}{T} \geq \frac{(n-1)^2}{n^2 - x^2}.$$

Together with (4.2), $|t| \leq \|\mu\|_2$, and (4.4), this yields

$$\|A^{-1}\|_F^2 \geq L(x) := \frac{4(n-1)^2}{n^2 - x^2} + \frac{4}{(n+x)^2}. \quad (4.5)$$

For $1 \leq x < n$, direct differentiation shows that

$$L'(x) \geq 0 \iff (n-1)^2 x(n+x) \geq (n-x)^2.$$

The last inequality is strict because its left-hand side is at least $(n-1)^2(n+1)$, while its right-hand side is at most $(n-1)^2$. Therefore

$$L(x) \geq L(1) = \frac{4(n-1)}{n+1} + \frac{4}{(n+1)^2} = \frac{4n^2}{(n+1)^2}. \quad (4.6)$$

It remains to rule out equality in the second case. If equality held in the desired estimate, then equality would have to hold at every step in (4.5)–(4.6). Since L is strictly increasing on $[1, n)$, this first gives $x = 1$. Equality in

$$T \leq n^2 - x^2$$

then gives $E = 0$ by (4.3); hence every entry of A is either 0 or 1. Equality in (4.4) forces $\mu - \mathbf{1}/2$ to be a nonnegative multiple of $\mathbf{1}$. Since $q = 1/(4n)$, it follows that

$$\mu = \frac{n+1}{2n}\mathbf{1}.$$

Thus every row of the binary matrix A would contain $(n+1)/2$ ones, which is impossible because n is even. Hence equality cannot occur. $\square$

The case $n = 2$ is handled directly.

Lemma 4.4. *If $A \in [0, 1]^{2 \times 2}$ is nonsingular, then*

$$\|A^{-1}\|_F^2 \geq 2 > \frac{16}{9}.$$

Proof. Put $F = \|A\|_F^2$. For a 2×2 matrix,

$$\|A^{-1}\|_F^2 = \frac{F}{(\det A)^2}.$$

If $F \leq 2$, Hadamard's determinant inequality [HJ13, p. 505, Theorem 7.8.1] and the arithmetic–geometric mean inequality applied to the squared row norms give

$$(\det A)^2 \leq \frac{F^2}{4} \leq \frac{F}{2}.$$

If $F \geq 2$, then $|\det A| \leq 1$, since $|a_{11}a_{22} - a_{12}a_{21}| \leq 1$, and again $(\det A)^2 \leq F/2$. The assertion follows. $\square$

Next, we prove Theorem 1.1.

Proof of Theorem 1.1. For odd n , including $n = 1$, the inequality and its equality statement follow from Proposition 2.3. For even $n \geq 4$, Proposition 4.3 gives strict inequality, while Lemma 4.4 handles $n = 2$. Finally, (1.5) shows that every S -matrix attains equality. These statements together prove the theorem. $\square$

Remark 4.5. For even $n \geq 2$, the proof gives strict inequality. For odd n , equality is attained exactly when a Hadamard matrix of order $n + 1$ exists. Theorem 1.1 therefore characterizes all possible equality cases without asserting existence in the unresolved Hadamard orders.

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